

IEEE 830 SRS document

Software Requirements Specification

for

StayEase – Hotel Management System

Version v5

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Revision History

Name	Date	Reason for Changes	Version
Initial version	24.04.2026	Initial version	v1
Second version	08.05.2026	Update to include the Use Case diagram and update some srs sections	v2
Update: add uml class diagrams	15.05.2026	Update to include UML class diagrams and update some srs sections, add user characteristics	v3
Update: add lab4 state diagrams and more srs changes	23.05.2026	Update to include sequence diagram and update External Interface Requirements and System Features/Modules sections	v4
Update: cleanup	03.06.2026	Clean up document, add missing parts	v5

1. Introduction

1.1. Purpose

The purpose of this document is to define the software requirements for the "StayEase" Hotel Management System. This SRS outlines the functional and non-functional requirements to ensure that the development team understands the system's scope, intended behavior, and quality constraints.

1.2. Document Conventions

This document follows the IEEE 830 standard. Requirement identifiers are prefixed with FR for Functional Requirements and NFR for Non-functional Requirements. The keyword shall is used to denote mandatory requirements. All text is formatted in a standard sans-serif font for readability.

1.3. Intended Audience and Reading Suggestions

This document is intended for developers, project managers, and hotel stakeholders. It is recommended that readers begin with the Product Scope and Problem Statement to understand the business context before reviewing the detailed functional requirements.

1.4. Product Scope

StayEase shall be a web-based management platform that shall automate room inventory and guest reservations. It shall replace manual scheduling with a synchronized digital environment, excluding hardware-level integrations like smart locks for this version.

2. Overall Description

2.1. Product Perspective and problem statement

Small and medium-sized hotels often face a dilemma when choosing management software. Enterprise cloud solutions frequently come with high recurring subscription fees that eat into the small margins of boutique businesses. Furthermore, these hotels are often located in areas where internet stability may fluctuate; a purely cloud-based system means that if the internet goes down, the hotel loses all access to its guest list, room status, and billing information.

Additionally, small hotel owners are increasingly concerned about data privacy and the security of their guests' personal information when stored on third-party servers. Storing sensitive guest details on a platform they do not control presents a significant long-term risk.

StayEase shall address these issues by being a self-hosted, full-stack application. By running the system on their own local infrastructure, hotel owners shall eliminate monthly subscription costs and maintain 100% ownership of their data. This architecture shall ensure that the "front of house" can continue to operate and access the "Single Source of Truth" during internet outages. StayEase shall provide small hotels with the professional tools of a large chain while respecting their need for autonomy, affordability, and data sovereignty.

2.2. Product Features

StayEase centralizes operations locally into four primary operational areas:

- **User Authentication & Security Control:** Secures local system access based on employee roles.
- **Room inventory Management:** Real-time digital allocation, filtering and automated updates of room cleanliness statuses.
- **Guest Reservation & booking engine:** creates and updates guest data, handling check-ins and conditional cancellations.

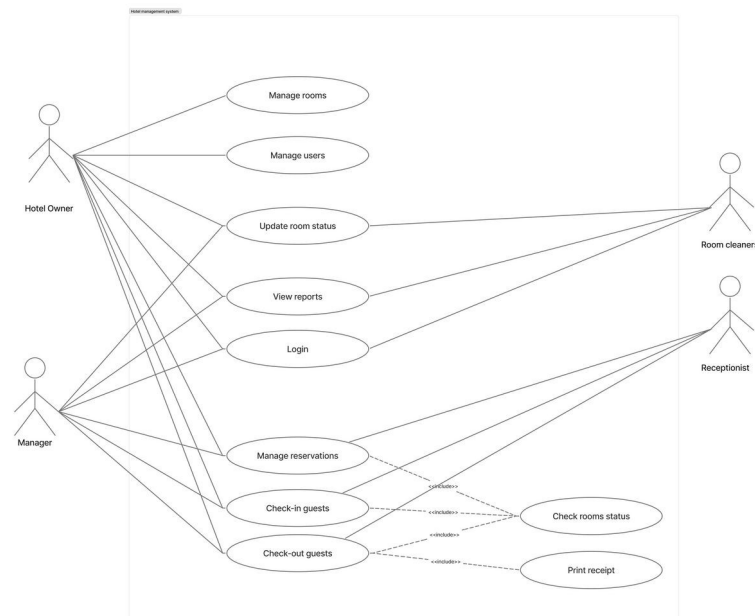
- Automated Billing Processing: Dynamically calculates client invoicing upon checkout based on localized parameters.

2.3. User Classes and Characteristics

The classes of users:

- Hotel Owner/Administrator – the frequency of use shall be sporadical, he shall have exclusive access to managing rooms and users, but he also should have access to all the other system features.
- Manager – the frequency of use shall be high, he shall have the second most permissions after hotel owner, he shall be able to update room status, view reports, manage reservations, check in guests and check out guests.
- Receptionist – the frequency of use shall be daily, and he should have access to managing reservations, as well as checking in guests and checking out guests.
- Room cleaners – the frequency of use might be daily, they should have access to updating room statuses and viewing reports/status of the rooms.

The basic use cases for the users are shown on the below diagram.



2.4. Operating Environment

The platform will be dedicated local computer acting as the server, it will have Ubuntu operation system in one of the latest distributions. As the database PostgreSQL will be used. Any device in the hotel, such as tablets, laptops or other workstations will need to be connected to the local network and by using modern browser, access the management website.

2.5. Design and Implementation Constraints

The system must be built as a self-hosted, full-stack web application designed to run locally on the client's own infrastructure.

The deployment environment is constrained to a dedicated local computer serving as the server running an Ubuntu operating system distribution.

Hardware-level ecosystem integrations, specifically including smart locks, are completely excluded from the scope of this system version.

The application database component is strictly constrained to PostgreSQL.

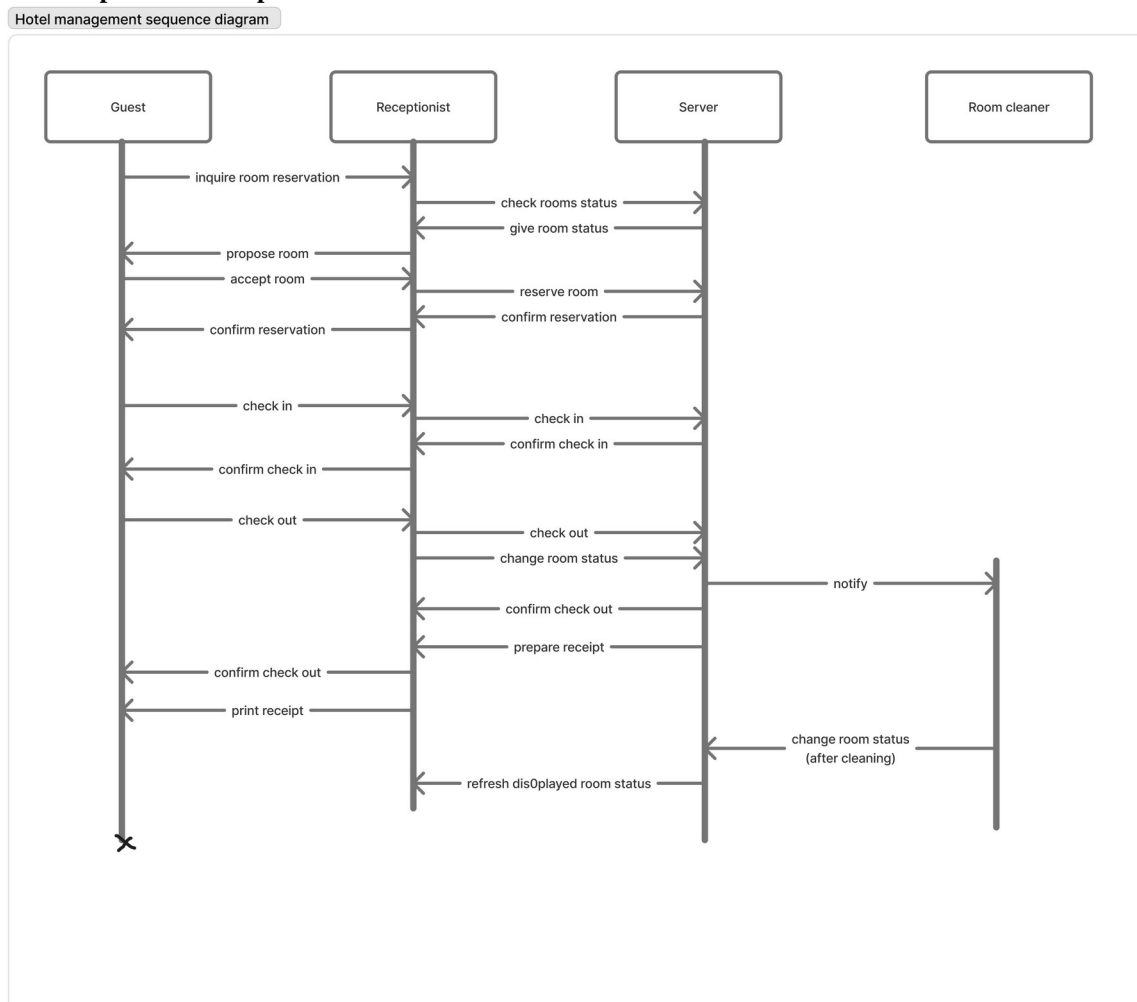
2.6. User Documentation

Staff will interact with a local management website interface.

User documentation components will be tailored around navigating the web-based system features across supported network devices.

User documentation will also be provided after logging in to the web page application.

2.7. Assumptions and Dependencies



It is assumed that all client interaction devices (tablets, laptops, or workstations) maintain a reliable connection to the hotel's local network environment.

The system relies on the local server computer remaining powered and active to ensure uninterrupted data sovereignty and front-of-house operations during external internet outages.

It is assumed that end-users will access the platform exclusively using up-to-date, modern web browsers.

3. System Features

3.1. User Roles

User Authentication & Security Control: Secures local system access dynamically based on designated employee roles.

3.2. Room management

Room Inventory Management: Provides real-time digital allocation, filtering, and automated updates of room cleanliness statuses.

3.3. Reservations

Guest Reservation & Booking Engine: Manages guest data entries, check-in operations, and conditional booking cancellations.

3.4. Billing calculation

Automated Billing Processing: Dynamically handles calculations for client invoicing upon checkout based on localized system parameters.

4. External Interface Requirements

4.1. User Interfaces Overview

StayEase will provide a fully web-based user interface accessed locally via standard web browsers.

The interface dynamically adapts features and view configurations depending on the logged-in user role (Hotel Owner, Manager, Receptionist, or Room Cleaners).

4.2. Hardware Interfaces

The application interfaces directly with the hardware resources of a dedicated local computer configured as the application server.

Client-side interactions occur over the local area network using end-user hardware, including tablets, laptops, and desktop workstations.

No physical interfaces are established with smart locks or alternative external hardware automation tools in this version.

4.3. Software Interfaces

The server application interfaces directly with the underlying Ubuntu operating system environment.

Data persistence interfaces directly with a local instance of the PostgreSQL database engine.

Client-side runtime execution environments are provided natively by modern web browsers.

5. System Features/Modules

5.1. User Authentication and Security Control

5.1.1 Description and Priority

This module secures local system access based on employee roles. It has a critical priority to uphold user data privacy and eliminate third-party security vulnerabilities.

5.1.2 Stimulus/Response Sequences

An employee submits credentials or form data via the system interface.

Response: The system validates the submission parameters and grants secure access matching the user's role criteria.

5.1.3 Functional Requirements

REQ-1.1: Upon form submission, the System shall validate that the password contains at least 8 characters.

REQ-1.2: Upon a search query, the System shall filter room inventory to display only items with the "Available" status flag.

REQ-1.3: Upon check-out, the System shall calculate the total bill by multiplying the "RoomRate" by the "StayDuration" (in nights).

REQ-1.4: Upon a cancellation request, the System shall change the booking status to "Cancelled" if the request is made > 24 hours before check-in.

REQ-1.5: upon guest check-out, the System shall update the room status to "Needs Cleaning" automatically.

5.2 Room inventory management

5.2.1 Description and Priority

Handles real-time digital allocation, filtering, and automated updates of room cleanliness statuses.

5.2.2 Stimulus/Response Sequences

A user initiates an inventory search query or finishes checking out a guest.

Response: The system filters the visible records or triggers an automated room cleanliness status update.

5.2.3 Functional requirements

REQ-2.1: Upon a search query, the System shall filter room inventory to display only items with the "Available" status flag.

REQ-2.2: Upon guest check-out, the System shall update the room status to "Needs Cleaning" automatically.

6. Nonfunctional Requirements

6.1. Performance

NF-1.1: The System shall return search results in less than 2.0 seconds under a load of 20 concurrent users.

6.2. Security

NF-2.1: The System shall encrypt all data sent between devices.

6.3. Reliability

NF-3.1 The System **shall** maintain an uptime of **99.9%** during the peak window of 08:00 to 22:00.